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| Inventor(s)          | Cok; Ronald S.  |

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### Method of making a stacked electronic component

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#### Abstract

A stacked electronic component comprises a stack of three or more print layers. Each print layer has an area less than any print layers beneath the print layer in the stack. Each print layer comprises a dielectric layer and a functional layer disposed on the dielectric layer. The functional layer comprises an exposed conductive portion that is not covered with a dielectric layer of any of the print layers and each exposed conductive portion is nonoverlapping with any other exposed conductive portion. A patterned electrode layer is coated on at least a portion of the stack and defines one or more electrodes. Each electrode of the one or more electrodes in electrical contact with an exclusive subset of the exposed conductive portions. The functional layers can be passive conductors forming capacitors, resistors, inductors, or antennas, or active layers forming electronic circuits.

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| <b>Inventors:</b> | <b>Cok; Ronald S. (Rochester, NY)</b>   |
| <b>Applicant:</b> | <b>X-Celeprint Limited (Dublin, IE)</b> |
| <b>Family ID:</b> | <b>82322381</b>                         |
| <b>Assignee:</b>  | <b>X-Celeprint Limited (Dublin, IE)</b> |
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*Primary Examiner:* Kim; Paul D

*Attorney, Agent or Firm:* Choate, Hall & Stewart LLP

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## Background/Summary

**PRIORITY APPLICATION** (1) This application is a continuation of U.S. patent application Ser. No. 17/146,295, filed on Jan. 11, 2021, now U.S. Pat. No. 11,490,519, the disclosure of which is hereby incorporated by reference herein in its entirety. **CROSS REFERENCE TO RELATED APPLICATIONS** (2) Reference is made to U.S. Pat. No. 10,050,351, entitled Multilayer Printed Capacitors, filed Jun. 18, 2015 by Bower et al and to U.S. Patent Application Publication No. 2018/0042110, entitled Printable 3D Electronic Structure, filed Aug. 3, 2017 by Cok, the disclosures of which are incorporated herein by reference in their entirety.

## TECHNICAL FIELD

(1) The present disclosure relates generally to structures and methods for making stacked electronic components using micro-transfer printing.

## BACKGROUND

(2) Electronic circuits typically use a combination of passive and active electronic components. Passive electronic components include capacitors, resistors, and inductors (e.g., coils) that affect the flow of current, store electrical charge, or form electromagnetic fields. Active electronic components include transistors and diodes incorporating semiconductor materials that can switch or amplify electronic current. For electronic systems comprising printed circuit boards, integrated circuits and stacked electronic components can be assembled on the printed circuit boards using methods such as pick-and-place. However, such techniques for assembling passive components are limited in the form factors that can be assembled due to limitations on the size of components that can be manipulated and placement accuracy.

(3) Micro-electronic systems can be highly integrated and include active micro-components with sizes of only a few nanometers integrated in a layer of a semiconductor substrate, for example as are found in integrated circuits useful in computers and portable electronic devices such as cellular telephones. Although the size of the active components can be very small, comparably sized passive components have much smaller functional parameters. For example, smaller resistors have a smaller resistance and smaller capacitors have a smaller capacitance. However, in many electronic circuits, even if the active components are reduced in size, the desired resistance or

capacitance of passive components in the electronic circuits remains the same. In consequence, the passive components can be relatively large compared to the active components, inhibiting a desired reduction in size of the electronic circuits and associated micro-electronic systems. There is a need, therefore, for smaller passive electronic components with larger functional parameters suitable for integration into micro-electronic devices such as integrated circuits. There is also a need for active electronic devices made in smaller structures.

(4) To increase density in electronic systems further, some electronic systems use stacked integrated circuits to reduce power, improve switching speed, and increase density, for example as taught in U.S. Pat. No. 6,551,857. However, these structures require packaged integrated circuits and thermal diffusion bond layers, increasing the size and interconnection complexity of the structure. Other methods use stacked die layers with through interconnects, for example as discussed in U.S. Pat. No. 9,000,577, but construction of through interconnections, for example with through silicon vias, is difficult and expensive. Other methods employ interface wafers with through silicon vias to interconnect bonded active-circuitry wafers (U.S. Pat. No. 8,129,256) or integrated circuits (U.S. Pat. No. 8,546,900) but these are limited in the number of layers that can be interconnected.

(5) Stacked capacitor structures are discussed in U.S. Pat. No. 5,864,177, but these are each interconnected with bond wires, use internal vias to connect the plates in each capacitor, and the layers are constructed using expensive photolithography. U.S. Patent Publication No. 2007/0290321 discloses chip and wire and flip-chip-compatible die stack capacitors (“stack caps”) that comprise separately fabricated multi-layer sections that are bonded together. Each stack cap is wire bonded to a host substrate and comprises conductive adhesive layers adhering two successively smaller single-layer capacitors (SLCs) each having a dielectric layer with metalized top and bottom surfaces. The metalized top surfaces are wire bonded to the host substrate, providing two power connections and a ground connection. However, wire bonding is a slow and relatively expensive connection process unsuited to mass production in large volumes, the number of functional layers in each stack cap is limited as is the number of stack caps, thereby limiting the capacitance of the die stack capacitor, and conductive adhesives can be less conductive than desired, reducing the performance and functional parameters of the die stack capacitor.

(6) A method for transferring active micro-components from one substrate to another is described in *AMOLED Displays using Transfer-Printed Integrated Circuits* published in the Proceedings of the 2009 Society for Information Display International Symposium Jun. 2-5, 2009, in San Antonio Tex., US, vol. 40, Book 2, ISSN 0009-0966X, paper 63.2 p. 947. In this approach, small integrated circuits are formed over a buried oxide layer on the process side of a crystalline wafer. The small integrated circuits, or chiplets, are released from the wafer by etching the buried oxide layer formed beneath the circuits. A PDMS stamp is pressed against the wafer and the process side of the chiplets is adhered to the stamp. The chiplets are pressed against a destination substrate or backplane coated with an adhesive and thereby adhered to the destination substrate. The adhesive is subsequently cured. In another example, U.S. Pat. No. 8,722,458 entitled *Optical Systems Fabricated by Printing-Based Assembly* teaches transferring light-emitting, light-sensing, or light-collecting semiconductor elements from a wafer substrate to a destination substrate or backplane.

(7) U.S. Pat. No. 10,050,351 discloses a micro-capacitor comprising stacked substrates assembled by micro-transfer printing. In embodiments of this design, multiple substrates of identical size are stacked in an offset configuration to expose electrical connections to each substrate in the stack. Each substrate can comprise an array of vertical capacitors in a substrate with a common connection. Wire bonds connect the plates to form a three-dimensional capacitor. However, wire bonding can be a relatively slow and cumbersome method for connecting electronic devices.

(8) U.S. Patent Publication No. 2018/0042110 discloses a stack of micro-devices that are electrically connected with connection posts. The micro-devices can be, among other things, capacitors but the micro-devices are relatively large compared to the capacitor itself and require

repeated photolithographic processing steps.

(9) There is a need, therefore, for structures and methods that enable electronic micro-components with increased functional parameters and reduced size that are suitable for incorporation into micro-electronic systems.

## SUMMARY

(10) The present disclosure provides, among other embodiments, structures, materials, and methods for a stacked electronic component comprising a stack of three or more print layers. Each print layer in the stack of three or more print layers has an area less than any of the print layers that are beneath the print layer in the stack. Each of the print layers comprises a dielectric layer and a functional layer disposed on the dielectric layer. The functional layer comprises an exposed conductive portion that is not covered with a dielectric layer of any of the print layers and the exposed conductive portion of each of the print layers is nonoverlapping with the exposed conductive portion of any other of the print layers. A patterned electrode layer is coated on at least a portion of the stack and defines one or more electrodes. Each electrode of the one or more electrodes is in electrical contact with an exclusive subset of the exposed conductive portions of each of an exclusive subset of the print layers. Multiple electrical connections to a common exposed conductive portion of a print layer are a common electrode. According to some embodiments, the stacked electronic component is a passive electronic component. According to some embodiments, the stacked electronic component is an active electronic component. In some configurations, the functional layer is exclusively an electrical conductor, exclusively a resistive electrical conductor, an epitaxial layer, or an active circuit. The stacked electronic component can be a multi-layer active circuit, a capacitor, a resistor, an inductor, or an antenna.

(11) According to some embodiments, each and every print layer of each of the stack of three or more print layers is electrically connected exclusively by the patterned electrode layer. According to some embodiments, less than all of the print layers of the stack of three or more print layers is electrically connected by the patterned electrode layer. The print layer of each of the stack of three or more print layers can be, but is not necessarily, adhered to an adjacent print layer with a layer of adhesive. The dielectric layer can be the layer of adhesive.

(12) In some configurations, the dielectric layer is a bottom dielectric layer and each print layer of the stack of three or more print layers comprises a top dielectric layer disposed on the functional layer on a side (surface) of the functional layer opposite the bottom dielectric layer. The top dielectric layer can cover the functional layer except for the exposed conductive portion. The top dielectric layer can be an inorganic dielectric or a layer of adhesive that adheres adjacent print layers of the stack of three or more print layers together.

(13) According to some embodiments, the patterned electrode layer defines only two electrodes. According to some embodiments, the one or more electrodes in the patterned electrode layer are disposed in a common layer. According to some embodiments, the one or more electrodes electrically connect fewer than all of the exposed conductive portions.

(14) According to some embodiments, the exposed conductive portions of at least two print layers of the stack of three or more print layers are electrically connected by one or more functional-layer connectors (e.g., functional-layer electrical connections) that are physically and spatially separate from the one or more electrodes and electrically separate from the one or more electrodes except through the functional layers (e.g., are not directly connected). The one or more functional-layer connectors can be disposed in a common layer, for example in a common coated and patterned metal layer disposed and patterned in common steps. The one or more functional-layer connectors can be disposed in a common layer with the one or more electrodes, for example in a common coated and patterned metal layer disposed and patterned in common steps. The one or more functional-layer connectors can electrically connect the functional layers in series or in parallel. The functional-layer connectors can electrically connect the exposed conductive portions of adjacent print layers of the stack of three or more print layers.

(15) According to some embodiments, the print layers of the stack of three or more print layers are shaped as a polygon, for example within the limits of a manufacturing process. In some embodiments, all of the exposed conductive portions of the print layers are disposed on opposing sides (opposing edges) of the polygon. In some embodiments, each of the exposed conductive portions of the print layers is disposed on three sides (three edges) of the polygon. In some embodiments, each of the exposed conductive portions of the print layers is disposed on one side (one edge) of the polygon. In some embodiments, all of the exposed conductive portions are disposed on one side (one edge) of the polygon. According to some embodiments, the print layers of the stack of three or more print layers are substantially rectangular or the dielectric layer has a rectangular shape with an extended tab and the exposed conductive portion is disposed on the tab. In some embodiments, none of the exposed conductive portions electrically connected to a first electrode of the one or more electrodes are disposed spatially between the exposed conductive portions electrically connected to a second electrode of the one or more electrodes different from the first electrode in a direction orthogonal to the stack. In some embodiments, some of the exposed conductive portions electrically connected to a first electrode of the one or more electrodes are disposed spatially between the exposed conductive portions electrically connected to a second electrode of the one or more electrodes different from the first electrode in a direction orthogonal to the stack.

(16) Each functional layer in the stack can be a passive electrical conductor that is a plate (e.g., a coated surface of a polygon filled with an electrical conductor), a serpentine conductor (e.g., a wire), or a conductor within and close to at least most of a perimeter of a polygon (e.g., a wire extending around most of, and near to the edge of, the polygon, for example closer to the perimeter than a center of the polygon). Each functional layer can be an epitaxial layer, for example a photolithographically processed epitaxial layer comprising electrically connected transistors or diodes, or both transistor and diodes. Each functional layer can be semiconductor substrate, for example a photolithographically processed semiconductor substrate comprising electrically connected transistors or diodes, or both.

(17) In some embodiments, a stacked electronic component of the present disclosure comprises an insulating layer disposed over the stack of three or more print layers and vias formed in the insulating layer. One or more electrodes can be electrically connected to the exposed conductive portions through the vias. In some embodiments, a stacked electronic component of the present disclosure is encapsulated, for example by an organic or inorganic dielectric. In some embodiments, a stacked electronic component of the present disclosure is planarized, for example by an organic or inorganic dielectric.

(18) According to some embodiments, a stacked electronic component comprises an alignment structure and the print layers of the stack of three or more print layers are one or more of adjacent to, aligned by, and in contact with the alignment structure. The alignment structure can align the stack and the print layers in one dimension or in two dimensions.

(19) In some embodiments, the functional layer of one or more print layers of the stack of three or more print layers is a vertical capacitor layer.

(20) In some embodiments, each print layer of the stack of three or more print layers comprises a broken (e.g., fractured) or separated tether. In some embodiments, the functional layer of one or more print layers of the stack of three or more print layers comprises a broken (e.g., fractured) or separated tether.

(21) According to some embodiments of the present disclosure, a stacked electronic component comprises a component substrate comprising a component print layer, the component print layer comprising a component dielectric layer and a component functional layer disposed on the component dielectric layer. Print layers are disposed on the component substrate. The component functional layer comprises a component substrate exposed conductive portion that is not covered with a dielectric layer of any of the print layers disposed on the component substrate and an

electrode of the one or more electrodes is in electrical contact with the component substrate exposed conductive portion. The component print layer can be a print layer in the stack of print layers.

(22) According to some embodiments, a stacked electronic component is a micro-component. An area of the stacked electronic component can be no greater than 40,000  $\mu\text{m}^2$ , a thickness of each print layer can be no greater than one micron, two microns, five microns, or 10 microns, and a thickness of the stack can be no greater than 5 microns, 10 microns, 20 microns, 30 microns, 50 microns, or 100 microns.

(23) The patterned electrode layer can cover no less than 5% of an area of the stacked electronic component, no less than 5% of an area of the stacked electronic component, no less than 10% of an area of the stacked electronic component, no less than 25% of an area of the stacked electronic component, no less than 50% of an area of the stacked electronic component, no less than 75% of an area of the stacked electronic component, no less than 85% of an area of the stacked electronic component. The area of the stacked electronic component can be the area in a horizontal direction orthogonal to a vertical direction of the stack of print layers.

(24) According to embodiments of the present disclosure, a method of making a stacked electronic component comprises providing one or more print-layer source wafers comprising print layers, providing a component source substrate, transfer printing a first print layer from a print-layer source wafer of the one or more print-layer source wafers onto the component source substrate, transfer printing a second print layer from a print-layer source wafer of the one or more print-layer source wafers onto the first print layer, transfer printing a third print layer from a print-layer source wafer of the one or more print-layer source wafers onto the second print layer thereby increasing the number of print layers in the stack of print layers, coating the stack with an electrode layer, and patterning the electrode layer to define one or more electrodes. Each electrode of the one or more electrodes can be in electrical contact with each exposed conductive portion of an exclusive subset of the print layers in the stack. Each print layer comprises a dielectric layer and a functional layer disposed on the dielectric layer. Each functional layer can comprise an exposed conductive portion that is not covered with a dielectric layer of any of the print layers, the second print layer has a smaller area than the first print layer and the third print layer has a smaller area than the second print layer, and each exposed conductive portion of the print layers in the stack is nonoverlapping with any other exposed conductive portion.

(25) Some embodiments comprise successively transfer printing print layers having successively smaller areas from a print-layer source wafer of the one or more print-layer source wafers onto the stack to increase the number of print layers in the stack. Each functional layer of each print layer comprises an exposed conductive portion that is not covered with a dielectric layer of any of the print layers. The one or more print-layer source wafers can be a single, common print-layer source wafer, a plurality of substantially identical print-layer source wafers, or a plurality of print-layer source wafers, at least some of which are different from each other. Some methods comprise rotating a print layer with respect to another different print layer while transfer printing the print layer onto or over the other print layer.

(26) The component substrate can be disposed on or in a component source wafer and methods of the present disclosure can comprise transfer printing the component to a target substrate. In some methods, the component substrate is a target substrate. Some methods comprise coating the stack with an insulating layer and forming vias in the insulating layer to expose the exposed conductive portions.

(27) Patterning the electrode layer can comprise forming one or more functional-layer connectors that each electrically connect the exposed conductive portions of at least two print layers in the stack and are physically and spatially separate from the one or more electrodes so that the one or more electrodes electrically connect fewer than all of the exposed conductive portions in the stack.

(28) According to some embodiments of the present disclosure, an active electronic component

comprises a stack of three or more print layers. Each print layer in the stack of three or more print layers has an area less than any of the print layers that are beneath the print layer in the stack. Each print layer in the stack of three or more print layers comprises a dielectric layer and an epitaxial layer disposed on or in the dielectric layer. The epitaxial layer can comprise an exposed conductive portion that is not covered with a dielectric layer of any of the print layers of the stack of three or more print layers and the exposed conductive portion of each of the print layers is nonoverlapping with the exposed conductive portion of any other of the print layers. A patterned electrode layer can be coated on at least a portion of the stack, the patterned electrode layer defining one or more electrodes, each electrode of the one or more electrodes in electrical contact with the exposed conductive portion of each of an exclusive subset of the print layers. The dielectric layer can be a layer of adhesive. The epitaxial layer can be a semiconductor substrate, or a layer of epitaxy disposed on (e.g., grown on) the dielectric layer. In some embodiments, the print layers in the stack of three or more print layers are active print layers.

(29) Some active-print-layer embodiments of the present disclosure can also comprise one or more passive print layers that each comprise a dielectric layer and a functional layer disposed on the dielectric layer. The functional layer comprises an electrical conductor, for example a patterned electrical conductor, and an exposed conductive portion that is not covered with a dielectric layer of any of the print layers in the stack of three or more print layers. Each exposed conductive portion is nonoverlapping with any other exposed conductive portion or any active print layer or passive print layer in the stack of three or more print layers.

(30) According to some embodiments, a passive electronic component comprises a stack of three or more print layers, each print layer in the stack having an area less than any of the print layers that are beneath the print layer in the stack of three or more print layers. Each of the print layers comprises a dielectric layer and a conductive layer disposed on or in the dielectric layer. The conductive layer can be as substantially conductive as the materials and processing methods for the conductive layer allow or can be a substantially resistive conductor with a desired resistance defined by the materials and processing methods for the conductive layer. The conductive layer comprises an exposed conductive portion that is not covered with a dielectric layer of any of the print layers in the stack of three or more print layers and the exposed conductive portion of each of the print layers is nonoverlapping with the exposed conductive portion of any other of the print layers. A patterned electrode layer is coated on at least a portion of the stack, the patterned electrode layer defines one or more electrodes, and each electrode of the one or more electrodes is in electrical contact with the exposed conductive portions of each of an exclusive subset of the print layers.

(31) The present invention provides, inter alia, structures and methods that enable the construction of passive or active electronic micro-components with a reduced footprint over a substrate and with increased functional parameters or circuit complexity and size. In certain embodiments, the assembly and electrical interconnection process for the passive electronic micro-components is simple and inexpensive requiring fewer process steps than known alternative methods and provides a robust, three-dimensional electronic structure that is expandable in a variety of configurations and circuits.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other objects, aspects, features, and advantages of the present disclosure will become more apparent and better understood by referring to the following description taken in conjunction with the accompanying drawings, in which:

(2) FIGS. 1A-1D are plan views and cross sections of different-area print layers of a micro-



assembled electronic component according to illustrative embodiments of the present disclosure;

(3) FIG. 1E is a plan view and FIG. 1F is a corresponding cross section of the print layers of FIGS. 1A-1D micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(4) FIG. 1G is a cross section of the structure of FIGS. 1E and 1F coated with an electrode layer according to illustrative embodiments of the present disclosure;

(5) FIG. 1H is a cross section of the structure of FIG. 1G with the electrode layer patterned to define electrodes according to illustrative embodiments of the present disclosure;

(6) FIG. 1I is a cut-away plan view corresponding of the cross section of FIG. 1H according to illustrative embodiments of the present disclosure;

(7) FIGS. 2A-2D are plan views and cross sections of different-area print layers of a micro-assembled electronic component according to illustrative embodiments of the present disclosure;

(8) FIG. 2E is a cut-away plan view of the print layers of FIGS. 2A-2D micro-assembled on a component substrate with patterned electrodes according to illustrative embodiments of the present disclosure;

(9) FIGS. 3A-3D are plan views and cross sections of different-area print layers of a micro-assembled electronic component according to illustrative embodiments of the present disclosure;

(10) FIG. 3E is a plan view and FIG. 3F is a corresponding cross section of the print layers of FIGS. 3A-3D micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(11) FIG. 3G is a cross section of the structure of FIGS. 3E and 3F coated with a dielectric according to illustrative embodiments of the present disclosure;

(12) FIG. 3H is a cross section of the structure of FIG. 3G with the dielectric patterned to form vias according to illustrative embodiments of the present disclosure;

(13) FIG. 3I is a cross section of the structure of FIG. 3H coated with an electrode layer according to illustrative embodiments of the present disclosure;

(14) FIG. 3J is a cross section and FIG. 3K is a corresponding cut-away plan view of the structure of FIG. 3I with the electrode layer patterned to form electrodes according to illustrative embodiments of the present disclosure;

(15) FIGS. 4A-4D are plan views and cross sections of different print layers of a micro-assembled electronic component according to illustrative embodiments of the present disclosure;

(16) FIG. 4E is a cross section and FIG. 4F is a corresponding cut-away plan view of the print layers of FIGS. 4A-4D micro-assembled on a component substrate with patterned electrodes according to illustrative embodiments of the present disclosure;

(17) FIGS. 5A-5D are plan views and cross sections of different print layers of an electronic component according to illustrative embodiments of the present disclosure;

(18) FIG. 5E is a cross section and FIG. 5F is a corresponding cut-away plan view of the print layers of FIGS. 5A-5D micro-assembled with an alignment structure on a component substrate and with patterned electrodes according to illustrative embodiments of the present disclosure;

(19) FIG. 6 is a cut-away plan view of a micro-assembled electronic component with patterned electrodes and an alignment structure micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(20) FIG. 7 is a cut-away plan view of a micro-assembled electronic component with patterned electrodes and a one-dimensional alignment structure micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(21) FIG. 8 is a cut-away plan view of a micro-assembled electronic component with patterned electrodes and a two-dimensional alignment structure micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(22) FIG. 9A is a cross section and FIG. 9B is a plan view of a printable vertical capacitor print layer according to illustrative embodiments of the present disclosure;

(23) FIG. 9C is a cut-away plan view of a micro-assembled stacked capacitor with vertical-capacitor print layers corresponding to FIGS. 9A and 9B and patterned electrodes micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(24) FIGS. 10A-10D are plan views and cross sections of different-area print layers of a resistor according to illustrative embodiments of the present disclosure;

(25) FIG. 10E is a cut-away plan view of a micro-assembled stacked resistor with layers corresponding to FIGS. 10A-10D and patterned electrodes micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(26) FIGS. 11A-11D are plan views and cross sections of different-area print layers of a micro-assembled resistor according to illustrative embodiments of the present disclosure;

(27) FIG. 11E is a cut-away plan view of a micro-assembled stacked resistor with print layers corresponding to FIGS. 11A-11D with patterned electrodes micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(28) FIGS. 12A-12D are plan views and cross sections of different-area print layers of a micro-assembled inductor or a micro-assembled antenna according to illustrative embodiments of the present disclosure;

(29) FIG. 12E is a cut-away plan view of a micro-assembled stacked inductor with print layers corresponding to FIGS. 12A-12D with patterned electrodes micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(30) FIG. 12F is a plan view of a micro-assembled stacked inductor with layers corresponding to FIGS. 12A-12E with patterned electrodes micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(31) FIG. 12G is a plan view of a micro-assembled stacked antenna with layers corresponding to FIGS. 12A-12D with a patterned electrode micro-assembled on a component substrate according to illustrative embodiments of the present disclosure;

(32) FIGS. 13-14 are cross sections of stacked electronic components comprising adhesive layers according to illustrative embodiments of the present disclosure;

(33) FIG. 15 is a cross section of a stacked electronic component on a target substrate according to illustrative embodiments of the present disclosure;

(34) FIG. 16A is a cross section of a stacked electronic component comprising a component substrate and a component functional layer on a dielectric layer with layers of adhesive according to illustrative embodiments of the present disclosure;

(35) FIG. 16B is a cross section of a stacked electronic component comprising a component substrate and a component functional layer with electrodes according to illustrative embodiments of the present disclosure;

(36) FIG. 16C is a cross section of a stacked electronic component comprising a component substrate and a component functional layer on a bottom adhesive dielectric layer with a top dielectric layer according to illustrative embodiments of the present disclosure;

(37) FIG. 16D is a cross section of a stacked electronic component comprising a component substrate and a component functional layer on an adhesive dielectric layer according to illustrative embodiments of the present disclosure;

(38) FIGS. 17 and 18 are flow diagrams illustrating methods of the present disclosure;

(39) FIG. 19 is a cross section of different-area transfer-printable printed layers in a print-layer source wafer according to illustrative embodiments of the present disclosure;

(40) FIG. 20 is a cross section of a transfer-printable stacked electronic component disposed on a component substrate in a component source wafer according to illustrative embodiments of the present disclosure;

(41) FIG. 21 is a plan view of different-area print layers with tabs of a micro-assembled component according to illustrative embodiments of the present disclosure;

(42) FIG. 22 is a plan view and cross section of a print layer having an epitaxial or semiconductor

layer and active circuit according to illustrative embodiments of the present disclosure; and (43) FIG. 23 is a cross section of a stacked electronic component disposed on and electrically connected to an active circuit in or on a target or component substrate according to illustrative embodiments of the present disclosure.

(44) The features and advantages of the present disclosure will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. The Figures are not drawn to scale since the variation in size of various elements in the Figures is too great to permit depiction to scale.

#### DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(45) The present disclosure provides, inter alia, structures and methods for stacked electronic components. The stacked electronic components can be active electronic components or passive electronic components. The stacked electronic components comprise three or more print layers disposed in a stack over a substrate, such as a component or a target substrate. The print layers are mutually non-native to each other, e.g., and any destination or target substrate but at least some print layers can be native to a common source wafer. Reference herein to “print layers” refers to, in some embodiments, three or more print layers. Each print layer comprises a dielectric layer (e.g., substrate) comprising dielectric material and a functional layer comprising electrically functional material. Each print layer in the stack has a successively smaller area than the previous print layer in the stack. The functional materials in each print layer are electrically connected together with a patterned coating of electrically conductive material, such as a metal, to form electrodes electrically connecting the functional layers to external devices. The functional materials in some print layers can be electrically connected together with functional-layer connections internal to the stack. The functional materials in the print layers can comprise, without limitation, passive electronic materials such as electrical conductors (e.g., planar, serpentine, or three dimensionally structured conductors, for example surrounded by dielectric that forms a planarizing layer) or active materials comprising epitaxial layers or semiconductor substrates comprising active circuits with transistors or diodes. The stacked electronic components can be passive electronic devices such as capacitors, resistors, inductors, and antennas or active electronic devices such as multi-layer integrated circuits.

(46) The stacked electronic components can be micro-assembled massively in parallel using transfer printing (e.g., micro-transfer printing) with excellent efficiency and at reduced cost in a simple, efficient, extensible, flexible, and cost-effective way. The stacked electronic components can be constructed in common processing steps with integrated circuits, can be very small, and can be integrated directly into or on integrated circuit dies or in unpackaged circuits on a micro-substrate such as a component substrate. Stacked passive electronic components of the present disclosure can provide, in a small, dense structure, functional parameters (e.g., capacitance, resistance, inductance) much greater than conventional thin-film planar structures found in integrated circuits.

(47) According to some embodiments of the present disclosure and with reference to FIGS. 1A-1I, a stacked electronic component **90** (e.g., a micro-assembled integrated circuit, a micro-assembled antenna **96**, a micro-assembled capacitor **97**, a micro-assembled resistor **98**, a micro-assembled inductor **99**) comprises a stack **80** of print layers **20**. As shown in FIGS. 1A-1D, each print layer **20** in stack **80** has an area less than any other print layers **20** beneath print layer **20** in stack **80**, so that stack **80** can have a pyramidal appearance with successively smaller layers toward the top of the pyramid, as shown in FIG. 1F. Stack **80** can extend in a vertical direction away from an underlying substrate, such as a component substrate **10**, so that an area of a print layer **20** is over a horizontal surface of component substrate **10** orthogonal to vertical direction D of stack **80**.

(48) Each print layer **20** comprises a dielectric layer **22** and a functional layer **24** disposed on

dielectric layer **22**. Dielectric layer **22** can be a dielectric substrate on which functional layer **24** is disposed (e.g., by micro-transfer printing so that functional layer **24** is non-native to dielectric layer **22**) or on which functional layer **24** is formed, so that functional layer **24** is native to dielectric layer **22**. Functional layer **24** can be a passive electrical conductor patterned to provide a specific function or an epitaxial layer or semiconductor substrate that can be processed to form an electronic circuit comprising active electronic devices, e.g., as found in integrated circuits.

(49) Functional layer **24** comprises an exposed conductive portion **25** (e.g., exposed conductive portions **25A** and **25B**, collectively exposed conductive portions **25**) that is not covered with a dielectric layer **22** of any of print layers **20**. Each exposed conductive portion **25** can be a passive electrical conductor such as a contact pad electrically connected to functional layer **24** and is spatially separated (e.g., by a spatial separation **S**) from any other exposed conductive portion **25** in a direction orthogonal to stack **80** (e.g., a horizontal direction **H**), as shown in FIG. **1E**. Stack **80** can be disposed on a component substrate **10** and extend from component substrate **10** in a vertical direction **D**, e.g., a vertical direction **D** orthogonal to horizontal directions **H** of a surface of component substrate **10**. Thus, a print layer **20** beneath another print layer **20** is vertically between the other print layer **20** and component substrate **10**. Likewise, a print layer **20** vertically above another print layer **20** is on or over a side (surface) of the other print layer **20** opposite component substrate **10**. Moreover, print layers **20** can be substantially planar (e.g., having a length and width in horizontal direction **H** much greater than a thickness in a vertical direction **D**) and extend in horizontal direction **H** orthogonal to vertical stack direction **D**. Spatial separation **S** can be in a horizontal direction **H** orthogonal to stack direction **D**. In some embodiments, spatial separation **S** is zero, that is two exposed conductive portions **25** can abut each other, but cannot overlap (e.g., are nonoverlapping) so that a vertical line extending from component substrate **10** in direction **D** cannot pass through two exposed conductive portions **25**.

(50) As shown in FIG. **1G**, an electrode layer **30** is coated on at least a portion of stack **80** and then patterned to define one or more electrodes **32** (e.g., electrodes **32A**, **32B**, collectively electrodes **32**), as shown in FIGS. **1H** and **1I**. Each electrode **32** is in electrical contact with an exclusive subset of exposed conductive portions **25**. Multiple electrical connections to a common exposed conductive portion **25** of a print layer **20** are a common electrode **32**. Coating and patterning electrodes **32** over stack **80** and in electrical contact with exposed conductive portions **25** provides a simple, efficiently constructed, and highly conductive connection to the multiple print layers **20** in stack **80**. In some embodiments, electrodes **32** are coated over print layer **20** surfaces (e.g., at least partially on dielectric layer **22**, including a vertical edge of dielectric layer **22**, and at least partially on exposed conductive portions **25**) and, optionally, at least partially on a top dielectric layer **26** disposed on or over at least a portion of functional layer **24**. In some embodiments, electrodes **32** and, more generally, stacked electronic components **90** do not comprise bond wires, which are fragile and prone to failure unless packaged (which greatly increases the component size).

(51) FIGS. **1A**, **1B**, **1C**, and **1D** illustrate four print layers **20** with successively smaller areas in plan view and cross section. In some embodiments, each print layer **20** comprises a dielectric layer **22** (for example an oxide or nitride layer such as silicon dioxide or silicon nitride or, in some embodiments, an organic resin, such as a cured polymer, e.g., epoxy or photoresist) and a functional layer **24** (for example a low-resistance electrical conductor such as a metal such as aluminum, copper, silver, or gold, a resistive electrical conductor such as polysilicon, or an epitaxial layer deposited by evaporation, chemical vapor deposition, or sputtering or a semiconductor substrate disposed by transfer printing, (e.g., micro-transfer printing) disposed and patterned on dielectric layer **22**, for example using photolithographic methods and materials. Print layers **20** are disposed on a substrate, for example component substrate **10**, in an ordered stack **80** so that each print layer **20** in stack **80** has an area smaller than all other print layers **20** beneath print layer **20** (e.g., closer to component substrate **10**). An area of a print layer **20** can be the area of print

layer **20** over component substrate **10** (e.g., a footprint of print layer **20**) equal to the product of an X dimension and a Y dimension of print layer **20** (shown in FIG. **1E** for print layer **20** of FIG. **1A**). A direction D of stack **80** (shown in FIG. **1F**) is a vertical upward (up) direction away from component substrate **10** in the direction of decreasing areas of print layers **20** in stack **80**. A top print layer **20** is the print layer **20** farthest from component substrate **10** and a bottom print layer **20** is the print layer **20** closest to component substrate **10**. A downward (down) direction is a vertical direction opposite D and in the direction of print layers **20** with increasing areas from top print layer **20** to bottom print layer **20**. Those knowledgeable in the art will understand that up and down, top and bottom, over and under, and above and beneath are relative references and can be exchanged, for example by turning stacked electronic component **90** over. Adjacent print layers **20** are print layers **20** for which no other print layer **20** is between the adjacent print layers **20** in direction D or opposite direction D. Bottom print layer **20** is adjacent to component substrate **10**.

(52) In embodiments according to FIGS. **1A-1I**, an additional dielectric layer (top dielectric layer **26**, for example comprising patterned oxides or nitrides such as silicon dioxide or silicon nitride or organic materials such as adhesive or epoxies) is disposed over functional layer **24** in each print layer **20**. In such embodiments, dielectric layer **22** can be a bottom dielectric layer **22**. Top dielectric layer **26** can have a top area (footprint) over component substrate **10** and bottom dielectric layer **22** can have a bottom area (footprint) over component substrate **10**. In each print layer **20**, the bottom area can be greater than an area of functional layer **24** and the area of functional layer **24** can be greater than the top area so that functional layer **24** extends beyond top dielectric layer **26** over bottom dielectric layer **22** to form exposed conductive portions **25A**, **25B**. Top dielectric layer **26** can cover all of functional layer **24** except exposed conductive portions **25**. According to some embodiments, each functional layer **24** is exclusively directly adjacent to a dielectric layer **22** in different print layers **20** or in the same print layer **20** (e.g., between bottom dielectric layer **22** and top dielectric layer **26**). Exposed conductive portions **25A** can be electrically connected to a common electrode **32A** and exposed conductive portions **25B** can be electrically connected to a common electrode **32B**. However, embodiments of the present disclosure are not limited to two electrodes **30** and two exclusive subsets of exposed conductive portions **25**. In some embodiments, three or more of each are provided. Top and bottom dielectric layers **26**, **22** and functional layers **24** can be constructed using photolithographic methods and materials, for example deposition by sputtering or evaporation and patterning by photoresist deposition, masking, and etching. In embodiments in which functional layer **24** is a semiconductor substrate, e.g., is an unpackaged micro-chip, the semiconductor substrate can be disposed on dielectric layer **22** by micro-transfer printing and can comprise a broken (e.g., fractured) or separated tether (not shown in the Figures).

(53) As shown in FIGS. **1E** and **1F**, print layers **20** are successively disposed on component substrate **10**, for example using transfer printing, such as micro-transfer printing, in order of area size, with the largest-area print layer **20** disposed first (e.g., print layer **20** as shown in FIG. **1A**) and the smallest-area print layer **20** disposed last (e.g., print layer **20** as shown in FIG. **1D**) to form stack **80** of print layers **20**. Each successive print layer **20** is disposed so that exposed conductive portion **25A** or **25B** on each print layer **20** remains exposed and extends beyond dielectric layers **22** of print layers **20** above it in stack **80**.

(54) For clarity, the plan view Figures of the present disclosure illustrate electrodes **32** (e.g., electrodes **32A** and **32B**) with transparent rectangles outlined with solid lines and top dielectric layers **26** with transparent rectangles outlined with dashed lines. Bottom dielectric layers **22** are illustrated with white rectangles outlined with solid lines and functional layers **24** are illustrated with filled rectangles outlined with solid lines. Exposed conductive portions **25** are electrically conductive portions of functional layers **24**, such as a contact pad. In some embodiments, print layers **20** are transfer printable print layers **20** with print-layer tethers **28**. For clarity of illustration, print-layer tethers **28** are not shown in the plan views and cross sections of print layer **20** stacks **80**.

(e.g., FIGS. 1E, 1I and FIGS. 1F-1H, respectively) but can be shown in the plan views of individual print layers **20** (e.g., FIGS. 1A-1D).

(55) Print layers **20** can be micro-transfer printed from a print-layer source wafer **70** (e.g., as shown in FIG. 19) comprising sacrificial portions **72** spatially and laterally separated by anchors **74** with a print layer **20** disposed completely, directly, and exclusively over each sacrificial portion **72** with a print-layer tether **28** physically connecting each print layer **20** to an anchor **74**. After sacrificial portion **72** is etched to form a gap between print layer **20** and print-layer source wafer **70**, a stamp, e.g., a visco-elastic stamp with stamp posts, contacts each print layer **20** to adhere print layer **20** to the stamp post, removes print layers **20** from print-layer source wafer **70** thereby fracturing print-layer tethers **28**, and disposes print layer **20** on stack **80** or component substrate **10**.

(56) Once print layers **20** are disposed in stack **80**, for example on or over component substrate **10**, electrode layer **30** can be disposed (e.g., coated by evaporation, sputtering, or spray coating), for example in an unpatterned blanket coating as shown in FIG. 1G and then patterned using photolithographic methods and materials. Electrode layer **30** can comprise any suitable electrical conductor, for example a metal such as aluminum, silver, gold, titanium, tungsten, copper, a metal alloy, a conductive oxide such as indium tin oxide, or a conductive polymer such as polythiophene. Electrode layer **30** can be, but is not necessarily, transparent or reflective. After deposition, electrode layer **30** is patterned, for example using photolithography, to form electrodes **32** (e.g., electrodes **32A**, **32B**) as shown in FIGS. 1H and 1I, to complete micro-assembled and stacked electronic component **90**. Thus, all of electrodes **32** are disposed, formed, and patterned in a common layer with common processing steps, reducing costs and improving manufacturing throughput.

(57) Electrodes **32** can be any coated and patterned electrical conductor, for example electrically conductive metal layers, can be metal oxide conductors, can be organic conductors such as polythiophene, can be transparent or opaque, and can be provided in various widths, materials, and thicknesses. In general, stacked electronic components **90** can have any number of electrodes **32**. For applications such as a micro-assembled antenna **96**, stacked electronic components **90** can have one or two electrodes **32**. For applications such as a micro-assembled capacitor **97**, micro-assembled resistor **98**, or micro-assembled inductor **99**, stacked electronic components **90** can have two electrodes **32**, for example only two electrodes **32**. For applications forming an active electronic circuit (e.g., a stacked integrated circuit), stacked electronic components **90** can have more than two electrodes **32**, for example three to ten electrodes **32**, and can include power, ground, and signal electrodes **32**. FIGS. 1H (in cross section) and 1I (in plan view) illustrate a stack **80** of print layers **20** with bottom dielectric layers **22**, functional layers **24**, patterned top dielectric layers **26**, and exposed conductive portions **25** (e.g., exposed conductive portions **25A** and **25B**) electrically connected with electrodes **32** (e.g., electrodes **32A**, **32B**) disposed on component substrate **10** with component tether **18**.

(58) Coated electrodes **32A**, **32B** can be made in parallel using photolithography for a great number of stacked electronic components **90** on a component source wafer **76** (e.g., as shown in FIG. 20) at the same time and are therefore made more efficiently than other interconnection methods such as bond wires, which are sequentially connected by a bond wire machine. Coated electrodes **32A**, **32B** can also have a greater conductivity and emit less electromagnetic radiation in operation than bond wires, since they can be thicker, curve less, and do not extend away from print layers **20** or component substrate **10** (e.g., into the air). Coated electrodes **32A**, **32B** can also have smaller dimensions than bond wires, for example having a width or thickness no greater than 25 microns, no greater than 20 microns, no greater than 15 microns, no greater than 10 microns, no greater than 5 microns, or no greater than 2 microns. In some embodiments, exposed conductive portions **25A**, **25B** of print layers **20** are electrically connected exclusively by patterned electrode layer **30** (e.g., by electrodes **32A**, **32B**) and are not electrically connected by bond wires. In some embodiments, exposed conductive portions **25** of print layers **20** can have an exposed length or width in a

horizontal direction H that is less than 25 microns, less than 20 microns, less than 15 microns, less than 12 microns, or less than 10 microns. Such small exposed conductive portions **25** can be too small to electrically connect with bond wires but are readily electrically connected with electrodes **32**. Moreover, embodiments of the present disclosure enable electrical connections between different print layers **20** in stack **80** without the use of expensive and complex through-silicon vias or through-substrate vias (e.g., vias through dielectric layer **22**), reducing costs and enhancing manufacturability of stacked electronic components **90** of the present disclosure.

(59) As shown in FIG. **20**, stacked electronic component **90** can be constructed on a component substrate **10** of a component source wafer **76** with sacrificial portions **72** laterally separated by anchors **74** and a stacked electronic component **90** disposed exclusively, directly, and completely over each sacrificial portion **72**, connected to anchor **74** with component tether **18**, and transfer printed to a target substrate **50** (shown in FIG. **15** discussed below) using a stamp to adhere stacked electronic component **90** to the stamp, removing the stamp with stacked electronic component **90** and fracturing component tether **18**, and printing stacked electronic component **90** to target substrate **50**. Once stacked electronic component **90** is disposed on its final substrate (e.g., formed in place on component substrate **10**, formed in place on target substrate **50**, or transfer printed from component source wafer **76** to target substrate **50**), it can be electrically interconnected to any electronic circuit or wires on final target substrate **50**, for example with metal wires constructed using photolithographic methods and materials. In some embodiments, the electrical connection of stacked electronic component **90** to any external circuits or electrical connections is done in a common step with disposing electrode layer **30** and patterning electrodes **32A**, **32B**, reducing photolithographic processing steps and costs for the final electronic system.

(60) Component substrate **10** or a final target substrate **50** can be a glass or polymer substrate or, in some embodiments, an unpackaged semiconductor die, for example a portion of an integrated circuit and for example part of a semiconductor wafer having a surface on which print layers **20** can be disposed, e.g., transfer printed. In some embodiments, component substrate **10** or a final target substrate **50** is a module substrate on which an unpackaged integrated circuit or semiconductor die is disposed. Such integrated circuits can be, but are not limited to, silicon circuits, such as CMOS, or compound semiconductor circuits formed in doped or undoped compound semiconductors such as GaN, GaAs, InP and comprising light-emitting diodes, high-power or high-electronic mobility transistors, micro-electromechanical device, and sensors. Component substrate **10** or final target substrate **50** can be, but is not limited to, glass, plastic, semiconductor, compound semiconductor, or ceramic. Generally, component substrate **10** or target substrate **50** can be any substrate on which print layers **20** and electrode layer **30** can be disposed, for example a semiconductor substrate or a glass or plastic substrate as found in the display or integrated circuit industries. Component substrate **10** can be rigid or flexible as well as transparent or opaque.

(61) In the FIGS. **1A-1I** illustrations, stacked electronic component **90** and print layers **20** are substantially rectangular (excluding any print-layer tethers **28** or component tethers **18**) and exposed conductive portions **25** are disposed on opposing sides (opposing edges) of the rectangle. More generally, stacked electronic component **90** and print layers **20** can have any shape in horizontal direction H over component substrate **10** or target substrate **50**, for example polygonal shapes. Exposed conductive portions **25** can be disposed on opposing sides (opposing edges) of a perimeter of the polygon. As shown in the embodiments of FIGS. **2A-2E**, stacked electronic component **90** and print layers **20** are substantially rectangular and exposed conductive portions **25** are each disposed on three sides of the rectangle, extending completely over opposing sides and less than halfway on the other sides to enable exclusive electrical connection to different electrodes **32**. (As used herein, when referring to a vertical direction D, a side of print layer **24** is a surface of print layer **24**, when referring to a shape of print layer **24** or horizontal direction H, a side of print layer **24** is an edge or portion of a perimeter of print layer **24** in a horizontal plane.)

(62) The different shapes of exposed conductive portions 25 can be defined by appropriately patterning top dielectric layer 26 or by patterning functional layer 24. FIGS. 2A-2D illustrate print layers 20 similar to those of FIGS. 1A-1D except with a differently patterned top dielectric layer 26 disposed over functional layer 24 on bottom dielectric layer 22. FIG. 2E illustrates a stack 80 of print layers 20 with bottom dielectric layers 22, functional layers 24, patterned top dielectric layers 26, and exposed conductive portions 25 (e.g., 25A and 25B) electrically connected with electrodes 32 (e.g., electrodes 32A, 32B) disposed on component substrate 10 with component tether 18. In some embodiments and as illustrated in FIGS. 4A-4F discussed further below, exposed conductive portions 25 can all be disposed on a common side (common edge) of stacked electronic component 90 and a common side of all of print layers 20. As with the embodiments of FIGS. 2A-2E, this can be achieved by appropriately patterning top dielectric layer 26.

(63) By appropriately patterning top dielectric layer 26 to expose desired portions of functional layer 24 (e.g., exposed conductive portions 25), exposed conductive portions 25 (e.g., exposed conductive portions 25A, 25B) can be spatially located over component substrate 10 and in stacked electronic component 90 to enable a simple pattern at low resolution for electrodes 32 electrically connecting exposed conductive portions 25A and 25B corresponding to electrodes 32A and electrode 32B, respectively. Moreover, larger or more extensive exposed conductive portions 25 can increase an electrical connection area to functional layer 24. Such simple electrode 32 shapes (e.g., rectangles) improve current flow, reduce electromagnetic radiation, and reduce resolution requirements for the masks needed to pattern electrodes 32. Such spatial locations can be achieved by disposing none of exposed conductive portions 25 electrically connected to a first electrode 32 spatially between exposed conductive portions 25 electrically connected to a different second electrode 32 in horizontal direction H (e.g., a direction orthogonal to stack direction D and parallel to a surface of component substrate 10). For example, and as shown in FIGS. 1H and 1I and in FIG. 2E, exposed conductive portions 25A are electrically connected to electrode 32A and exposed conductive portions 25B are electrically connected to electrode 32B. None of exposed conductive portions 25A are spatially located in horizontal direction H between any of exposed conductive portions 25B. Likewise, none of exposed conductive portions 25B are spatially located in a horizontal direction between any of exposed conductive portions 25A.

(64) In embodiments in which electrodes 32 have more complex shapes, exposed conductive portions 25A electrically connected by electrodes 32A can be disposed spatially in horizontal direction H between exposed conductive portions 25B electrically connected by electrodes 32B.

(65) In embodiments according to FIGS. 1A-1I and 2A-2E, a top dielectric layer 26 is disposed over functional layer 24 in each print layer 20. Top dielectric layer 26 provides environmental protection to functional layer 24 but require an extra deposition and patterning step. In some embodiments, print layers 20 do not comprise a top dielectric layer 26. As shown in the embodiments of FIGS. 3A-3D, different-area print layers 20 each comprise a dielectric layer 22 with a functional layer 24 patterned on dielectric layer 22. Different-area print layers 20 are disposed in stack 80 on component substrate 10, for example by micro-transfer printing, as shown in the FIG. 3E plan view and FIG. 3F cross section with exposed conductive portion 25 of functional layer 24 in each print layer 20 exposed around or adjacent to the perimeter of each print layer 20. An unpatterned insulating layer 40, such as an uncured resin, adhesive, or photoresist, can be coated over stack 80 of print layers 20, as shown in FIG. 3G, and then patterned (for example using photolithographic patterning methods and materials) to form a patterned insulating layer 42 with vias 44 exposing exposed conductive portion 25 of each print layer 20, as shown in FIG. 3H. An unpatterned electrode layer 30 is then disposed over patterned insulating layer 42, vias 44, and exposed conductive portions 25, as shown in FIG. 3I, and patterned (for example using photolithographic methods and materials) to form electrodes 32A and 32B, as shown in FIGS. 3J and 3K. Thus, some embodiments of the present disclosure comprise a stacked electronic component 90 comprising a patterned insulating layer 42 disposed over stack 80 of print layers 20,



vias **44** formed in patterned insulating layer **42**, and one or more electrodes **32** electrically connected to exposed conductive portions **25** through vias **44**.

(66) As shown in FIGS. **4A-4F** and according to some embodiments of the present disclosure, exposed conductive portions **25** can be arranged without the use of a patterned insulating layer **42** as in FIGS. **3A-3K**. FIGS. **4A-4D** illustrate different-area print layers **20** comprising a dielectric layer **22** on each of which is patterned a functional layer **24**. Functional layers **24** and exposed conductive portions **25** of successive print layers **20** in stack **80** are alternately offset on the surface of and with respect to dielectric layer **22**. The different print layers **20** are transfer printed into a stack **80** and patterned with a conductor (e.g., a metal), as shown in FIGS. **4E** and **4F**, to form electrodes **32A** and **32B** in a stacked electronic device **90**. Dielectric layer **22** in each print layer **20** in stack **80** covers functional layer **24** of a print layer **20** beneath it in stack **80** except for the desired exposed conductive portion **25**. Functional layers **24** offset with respect to dielectric layers **22** in print layers **20** enable stacked electronic components **90** without top dielectric layers **26**, reducing the number of deposition and patterning steps necessary to construct stacked electronic components **90**.

(67) Any manufacturing process has resolution and alignment limitations. Transfer printing processes with large stamps and many print layers **20** (e.g., ten to one hundred thousand print layers **20**) can have a print accuracy of 1-2 microns on target substrate **50** and in small transfers in optimal conditions a print accuracy of less than one micron, for example several hundred nanometers. Since stack **80** can comprise many print layers **20** and allowance must be made for the accuracy of each print step in any practical manufacturing process, according to some embodiments of the present disclosure and as illustrated in FIGS. **5A-8**, a stacked electronic component **90** can comprise or be aligned with an alignment structure **12**, for example disposed on component substrate **10** or target substrate **50**, that can assist in aligning print layers **20**. As shown in FIGS. **5A-5D**, different-area print layers **20** each comprise a functional layer **24** patterned on a bottom dielectric layer **22** with an exposed conductive portion **25** (e.g., exposed conductive portions **25A**, **25B**) extending beyond patterned top dielectric layer **26**. As shown in the cross section of FIG. **5E** and cut-away plan view of FIG. **5F**, print layers **20** with successively smaller areas, each comprising bottom dielectric layers **22**, functional layers **24** on bottom dielectric layers **22**, and top dielectric layers **26** on functional layers **24**, are disposed in a stack **80** adjacent to, aligned by, and/or in contact with alignment structure **12**. Alignment structure **12** has a side adjacent to a corresponding side of print layers **20** so that print layers **20** are aligned in one dimension along a side of alignment structure **12**. In some embodiments and as shown in FIG. **5F**, print layers **20** aligned with alignment structure **12** have exposed conductive portions **25** (e.g., exposed conductive portions **25A**, **25B**) on a common side (or edge) of print layers **20** opposite alignment structure **12**. Exposed conductive portions **25A** electrically connected to electrode **32A** are on one half of the common side (edge) of print layers **20** and exposed conductive portions **25B** electrically connected to electrode **32B** are on the other half of the common side (edge) of print layers **20**. Such an arrangement enables a simple structure (e.g., a rectangular structure) for electrodes **32**.

(68) In some embodiments and as shown in FIG. **6**, exposed conductive portions **25** can extend onto two sides (edges) of print layers **20**. As shown in FIG. **7**, exposed conductive portions **25A** and **25B** can be on opposing sides (edge) of print layers **20**. FIG. **8** illustrates embodiments in which print layers **20** are aligned in two dimensions adjacent to, aligned by, and/or in contact with alignment structure **12** (or multiple orthogonal alignment structures **12**) with exposed conductive portions **25** on a common side (edge) of print layers **20**. Although alignment structures **12** are shown as solid and continuous along print layer **20** edges, in some embodiments alignment structures **12** extend only along a portion of print layer **20** edges.

(69) According to some embodiments of the present disclosure, print layers **20** in stack **80** have successively smaller areas towards the top of stack **80** opposite component substrate **10** or target substrate **50**. Print layers **20** can be arranged in a pyramid, so that edges on all sides of each print

layer **20** are exposed, for example as shown in FIGS. **1A-4F**. In some embodiments, common edges are aligned, for example one edge of each print layer **20** each respective edge in a common plane orthogonal to a surface of print layers **20**, as shown in FIGS. **5E-7**, or two edges of each print layer **20** each respective edge in a common plane orthogonal to a surface of print layers **20**, as shown in FIG. **8**. In some embodiments, not shown in the Figures, more than two edges of print layers **20** are aligned, for example three edges of rectangular print layers.

(70) In general, according to some embodiments of the present disclosure, there is no particular limitation on the arrangements or locations of exposed conductive portions **25** of the different-area print layers **20**, so long as an electrode **32** can electrically connect exposed conductive portions **25** of a subset of print layers **20**, e.g., an exclusive subset of print layers **20**. Electrodes **32** can have a variety of shapes including rectangles, curves, serpentine, and irregular configurations.

(71) Embodiments of the present disclosure can enable a variety of stacked passive electronic components **90**, including micro-assembled capacitors **97**, micro-assembled resistors **98**, micro-assembled inductors **99**, and micro-assembled antennas **96** (discussed further below), and print layers **20** in each of these can have a variety of configurations. For example, as illustrated in FIG. **1A-8**, functional layer **24** of each print layer **20** can be a conductive plate in a micro-assembled capacitor **97**. As illustrated in FIGS. **9A-9C**, print layers **20** can each be a vertical capacitor. A vertical capacitor can comprise one or more wells (e.g., holes, pits, or recesses) formed in dielectric layer **22**, for example by etching with photolithographic methods and materials. A first functional layer **24A** is disposed on dielectric layer **22** and in the wells. A patterned insulating layer **42** is disposed over first functional layer **24A**, leaving conductive portion **25A** exposed. A second functional layer **24B** is disposed on patterned insulating layer **42** and in the wells and is optionally coated with top dielectric layer **26**, leaving conductive portion **25B** exposed. This vertical capacitor structure provides increased capacitance to print layer **20**.

(72) FIGS. **1A-9C** illustrate various configurations of a passive electronic micro-assembled capacitor **97** comprising stacks **80** of print layers **20**. In these configurations, print layers **20** form two exclusive subsets with their exposed conductive portions **25** that include all of print layers **20**. Each exclusive subset is electrically connected in parallel. According to some embodiments of the present disclosure and as illustrated in FIGS. **10A-11E**, stacks **80** of print layers **20** form passive micro-assembled resistors **98**. FIGS. **10A-10E** illustrate micro-assembled resistor **98** embodiments comprising top dielectric layer **26** and FIGS. **11A-11E** illustrate micro-assembled resistor **98** embodiments that do not comprise top dielectric layer **26** (e.g., corresponding to micro-assembled capacitors **97** of FIGS. **1A-1I** and **3A-3K**, respectively). As illustrated in FIGS. **10A-10D**, print layers **20** each comprise bottom dielectric layer **22**, functional layer **24** patterned on bottom dielectric layer **22**, and top dielectric layer **26** patterned on functional layer **24**, leaving exposed conductive portions **25**. As illustrated in FIGS. **11A-11D**, print layers **20** each comprise dielectric layer **22** and functional layer **24** patterned on dielectric layer **22** leaving exposed conductive portions **25**. In both sets of embodiments, functional layer **24** forms an extended resistive electrical conductor, for example in a serpentine shape or other extended line and, for example, comprising polysilicon or a high-resistance metal or metal alloy. Exposed conductive portions **25A**, **25B** are at either end of the resistive electrical conductor in each print layer **20**. As shown in FIGS. **10E** and **11E**, exposed conductive portion **25A** of one print layer **20** is electrically connected to electrode **32A** and exposed conductive portion **25B** of one print layer **20** is electrically connected to electrode **32B**. The remaining print layers **20** are electrically connected in series by patterned functional-layer connectors **34** to form a single, multi-layer micro-assembled resistor **98**.

(73) According to some embodiments of the present disclosure and as illustrated in FIGS. **12A-12F**, a stacked electronic component **90** can be constructed as a micro-assembled inductor **99**. As shown in FIGS. **12-12D**, different-area print layers **20** each have a patterned functional layer **24** disposed on a bottom dielectric layer **22** and covered with an optional top dielectric layer **26**. Functional layers **24** are each patterned to form a conductive line (wire) that extends substantially around and

within the perimeter of dielectric layer **22**, but the ends of the conductive line in a print layer **20** are not electrically connected. FIG. **12E** illustrates a cut-away plan view showing the conductive lines of functional layers **24** and FIG. **12F** is a plan view showing the structure with top dielectric layers **26** covering functional layers **24**. As with micro-assembled resistor **98**, functional-layer connectors **34** serially electrically connect exposed conductive portions **25** of adjacent print layers **20** so that the conductive lines form a spiral of increasing height (in this case with substantially rectangular sides corresponding to rectangular print layers **20**). In some embodiments (not shown), each functional layer **24** in each print layer **20** can comprise a spiral that extends from an edge to a center and then, with additional metal layers passing over print layer **20** spiral to an edge where it is connected with functional-layer connector **34**. A spiral formed in each print layer **20** can increase the inductance of micro-assembled inductor **99**. Exposed conductive portions **25A**, **25B** at opposite ends of the spiral conductive lines electrically connected with functional-layer connectors **34** are electrically connected with electrodes **32A**, **32B** to complete micro-assembled inductor **99**.

(74) In some embodiments, the same micro-assembled inductor **99** can be a micro-assembled antenna **96** and can be connected at both ends of serially connected conductive functional layers **24** (e.g., by electrodes **32A**, **32B** as shown in FIG. **12F**), or at only one end of serially connected functional layers **24** (e.g., by electrode **32A** as shown in FIG. **12G**).

(75) Patterned functional-layer connectors **34** can be formed in a common step with common materials in a common layer (e.g., photolithographically defined metal layers) and are physically similar to electrodes **32**. Electrodes **32** electrically connect exclusive subsets of print layers **20** to external circuits that can use stacked electronic components **90**. Patterned functional-layer connectors **34**, in contrast, are electrical connections between print layers **20** (e.g., between exposed conductive portions **25** of different print layers **20**) in stack **80** and are not externally electrically connected. Because patterned functional-layer connectors **34** and electrodes **32** can be formed together in a common layer deposition and patterning process, they can be made efficiently and at relatively low cost. Thus, in some embodiments, each electrode **32** is electrically connected to an exposed conductive portion **25** of only one print layer **20** and, more generally, electrodes **32** electrically connect fewer than all of exposed conductive portions **25** of print layers **20**. Moreover, in some embodiments, exposed conductive portions **25** of at least two print layers **20** are electrically connected by one or more patterned functional-layer connectors **34** that are electrically separate from one or more electrodes **32**. Thus, functional-layer connectors **34** can electrically connect functional layers **24** in series and each functional-layer connector **34** can, but do not necessarily, directly electrically connect exposed conductive portions **25** of adjacent print layers **20**. In some embodiments and as shown in FIGS. **10A-12F**, electrode **32A** is electrically connected to bottom print layer **20** (e.g., print layer **20** with the largest area at the bottom of stack **80**) and electrode **32B** is electrically connected to top print layer **20** (e.g., print layer **20** with the smallest area at the top of stack **80**).

(76) Functional layer **24** can be patterned in a variety of patterns corresponding to the desired functionality of stacked electronic component **90**. For capacitive applications, each functional layer **24** can be a plate such as a polygon having an area filled with low-resistance conductive material, such as a metal, suitable for holding a charge and forming an electric or magnetic field between print layers **20**. For resistive applications, each functional layer **24** can be a serpentine line or wire comprising a resistive material, such as polysilicon or a high-resistance metal or electrical conductor, suitable for conducting electrical current with a useful and defined resistance. Each end of the serpentine line can be disposed at an edge of dielectric layer **22** and electrically connected with functional layer connectors **34**. For micro-assembled inductive or micro-assembled antenna **96** applications, each functional layer **24** can be a line or wire comprising a low-resistance conductive material, such as a metal, suitable for forming electrical or magnetic fields or responding to or forming electromagnetic radiation. The lines can be disposed around the perimeter of dielectric layer **22** (e.g., a polygon) and form one or more turns of a spiral in each print layer **20**. In active

circuits, functional layer **24** can comprise a layer of epitaxy, e.g., a semiconductor, or a semiconductor substrate patterned to form the active circuit, e.g., comprising transistors or diodes. (77) According to some embodiments of the present disclosure, print layers **20** are disposed (e.g., micro-transfer printed) onto component substrate **10** or onto another print layer **20** without an adhesive coated on component substrate **10** or other print layer **20**, for example as shown in FIGS. **1A-12G**. According to some embodiments and as illustrated in FIGS. **13-15**, a layer of adhesive **60** can be disposed on component substrate **10** or print layer **20** before print layer **20** is transferred thereon. Adhesive **60** can be a resin, photoresist, an epoxy, or other adhesive, for example a polymer adhesive and can be curable, for example with heat or radiation, such as ultra-violet radiation. Adhesive **60** can be a non-conductive adhesive **60**. Adhesive **60** can be disposed as a patterned layer (for example using an inkjet printer) or an unpatterned layer (for example using spray, curtain, or spin coating). The layer of adhesive **60** can be very thin, for example some nanometers or tens of nanometers thick, for example 30 nm or 50 nm thick. Adhesive **60** can be patterned, for example using photolithographic methods and materials.

(78) FIG. **13** illustrates embodiments in which a layer of adhesive **60** is disposed between a first print layer **20** and component substrate **10** (or target substrate **50**) and between each print layer **20**, for example by spray coating. Print layers **20** of FIG. **13** comprise both bottom and top dielectric layer **22**, **26** so that adhesive **60** adheres bottom dielectric layer **22** of a first print layer **20** to component substrate **10** (or target substrate **50**) and top dielectric layer **26** of first print layer **20** to bottom dielectric layer **22** of a second print layer **20** adhered to first print layer **20** in a stack **80**.

(79) FIG. **14** illustrates embodiments in which a layer of adhesive **60** is disposed between a first print layer **20** and component substrate **10** and between each print layer **20**. Print layers **20** of FIG. **14** comprise only a single dielectric layer **22** (e.g., no top dielectric layer **26** is present) so that adhesive **60** adheres dielectric layer **22** of a first print layer **20** to component substrate **10** and adheres functional layer **24** of first print layer **20** to dielectric layer **22** of an adjacent second print layer **20** in a stack **80**. In such embodiments, if the layer of adhesive **60** is non-conductive, it can serve as top dielectric layer **26** for print layers **20**, so that functional layers **24** are protected with a dielectric protection layer. Thus, adhesive **60** can be top dielectric layer **26** in print layer **20**. In any case, dielectric layer **22** of second print layer **20** can protect functional layer **24** of first print layer **20**, as second print layer **20** is disposed over first print layer **20**.

(80) In some embodiments, adhesive **60** can be sprayed over stack **80** after each print step for each print layer **20** without patterning. Adhesive **60** will accumulate with each repeated spraying until the final print layer **20** is disposed on the top of stack **80**. The adhesive can then be cured, if desired, and exposed adhesive **60** removed. Thus, each print layer **20** in stack **80** is adhered to an adjacent print layer **20** with a non-conductive adhesive **60**. Adhesive **60** removal can be a simple single unpatterned etch (e.g., a wet etch or dry etch, such as a plasma etch) that effectively removes exposed adhesive **60**, leaving adhesive **60** between print layers **20** and component substrate **10** in place, and cleans exposed conductive portions **25**. After adhesive **60** is removed, electrode layer **30** material can be disposed, e.g., by evaporation or sputtering, in an unpatterned layer and then patterned to form electrodes **32** using photolithographic patterning methods and materials.

(81) As shown in the illustration of FIG. **15**, according to some embodiments of the present disclosure, a stacked electronic component **90** can comprise a component substrate **10**. Print layers **20** are disposed (e.g., by micro-transfer printing) in a stack **80** on component substrate **10**. Component substrate **10**, together with stack **80** of print layers **20** can be a micro-transfer printable component and comprise a component tether **18** (e.g., micro-assembled on component source wafer **76** as shown in FIG. **20**). As shown in FIG. **15**, component electrodes **36** can be disposed on component substrate **10** and electrically connect to electrodes **32** of stacked electronic component **90**. Micro-assembled stacked electronic component **90** can be transfer printed onto target substrate **50**, fracturing or separating component tether **18**. Stacked electronic component **90** can be electrically connected to electrical devices or wires on target substrate **50**.

(82) In some embodiments of the present disclosure and as shown in FIG. 16A, component substrate **10** can comprise a component functional layer **14** similar to functional layer **24** disposed on a component dielectric layer **16**. Component functional layer **14** can comprise a component exposed conductive portion **25** that is not covered with a dielectric layer **22** of any of print layers **20** in stack **80**. Thus, component substrate **10** can also be a print layer **20** and can form bottom print layer **20** of stack **80** so that component function layer **14** is also a functional layer **24** and component dielectric layer **16** is also dielectric layer **22** in a print layer **20**. FIG. 16A illustrates a layer of adhesive **60** between print layers **20** and component functional layer **14**. FIG. 16B illustrates a stack **80** of print layers **20** without layers of adhesive **60** on target substrate **50**.

Electrodes **32** and component electrode **36** electrically connect the exposed conductive portions **25**. (83) As noted with respect to FIG. 14, layers of adhesive **60** can be top dielectric layer **26**. As shown in FIG. 16C, layers of adhesive **60** can be bottom dielectric layer **22**. This can be useful in cases in which functional layer **24** is an epitaxial or semiconductor substrate layer with a top dielectric layer **26** disposed on the semiconductor substrate (functional layer **24**). The semiconductor substrate and top dielectric layer **26** can be transfer printed onto a layer of adhesive **60**. Top dielectric layer **26** and the semiconductor substrate can be a mask for an otherwise unpatterned etch to remove adhesive **60** to expose conductive portions **25**. As shown in FIG. 16D, layers of adhesive **60** can be bottom dielectric layer **22** and top dielectric layer **26**. Again, this arrangement can be useful in cases in which functional layer **24** is an epitaxial or semiconductor substrate layer. However, in these embodiments, an external mask is useful to pattern-wise remove adhesive **60** to expose conductive portions **25** and leaving adhesive **60** remaining in place to protect portions of functional layer **24** (e.g., semiconductor substrate) from undesired electrical connection to electrodes **32**.

(84) According to embodiments of the present disclosure, functional layers **24** can comprise metals, such as copper or aluminum, or metal alloys, and can have a thickness of 100 nm to 5 microns, for example 500 nm to one micron. Dielectric layers **22** can have a thickness of one micron to 10 microns, for example 2-5 microns and can be an inorganic dielectric, such as silicon dioxide or silicon nitride or can be an organic dielectric with a thickness less than one micron. Top dielectric layers **26** can have a thickness less than the thickness of bottom dielectric layers **22**, for example having a thickness of one micron to 4 microns, and can be an inorganic dielectric, such as silicon dioxide or silicon nitride, or an organic dielectric, such as a polymer or adhesive with a thickness less than one micron.

(85) Embodiments of the present disclosure provide stacked electronic components **90** with relatively large functional parameters constructed with a simple manufacturing process and reduced resolution requirements. Electrodes **32** can be relatively thick, have good electrical conductivity, and can have a relatively low resolution. Patterned electrode layer **30** (e.g., electrodes **32**) can cover no less than 5%, 10%, 20%, 50%, 70%, 80%, or 90% of an area of stacked electronic component **90**, providing excellent conductivity and thermal dissipation. Stacked electronic components **90** can be micro-components, for example having an area no greater than 40,000  $\mu\text{m}^2$  (e.g., 200 by 200 microns), no greater than 10,000  $\mu\text{m}^2$  (e.g., 100 by 100 microns), 2,500  $\mu\text{m}^2$  (e.g., 50 by 50 microns), or 400  $\mu\text{m}^2$  (e.g., 20 by 20 microns). Since each print layer **20** can be relatively thin (e.g., 1 micron, 2 microns, 5 microns, or 10 microns), stack **80** can comprise 4 or more print layers **20**, 8 or more print layers **20**, 12 or more print layers **20**, 15 or more print layers **20**, 20 or more print layers **20**, or 50 or more print layers **20**, and a thickness of each print layer **20** can be no greater than 2 microns, no greater than 5 microns, or no greater than 10 microns. A thickness of stack **80** can be no greater than 10 microns, no greater than 15 microns, no greater than 20 microns, no greater than 50 microns, or no greater than 100 microns.

(86) As shown in FIG. 17, according to the present disclosure, methods of making a stacked electronic component **90** comprise providing one or more print-layer source wafers **70** comprising print layers **20** in step **100**, for example as shown in FIGS. 1A-1D and 19, and providing a

component source substrate **76** (e.g., a component source wafer **76**) in step **102** and as shown in FIG. **20**. Print layers **20** on a print-layer source wafer **70** can have a common area or have different areas as shown in FIGS. **1A-1D** and **19**. Thus, a single print-layer source wafer **70** can be a single, common print-layer source wafer **70** for all print layers **20**, or multiple different or same print-layer source wafers **70** can be used. Optionally, a layer of adhesive **60** is disposed on component source substrate **76** in step **106**. A first print layer **20** is transfer printed from one of print-layer source wafers **70** onto component source substrate **76** in step **110**. Optionally, in step **115**, adhesive **60** is disposed on first print layer **20**. In step **120**, a second print layer **20** is transfer printed from a print-layer source wafer **70** (e.g., the same print-layer source wafer **70** that provided first print layer **20** or a different print-layer source wafer **70**) onto the first print layer **20** to form a stack **80** of print layers **20**. Second print layer **20** has a smaller area over component source substrate **76** than first print layer **20**. In embodiments, second print layer **20** is completely and entirely disposed directly over first print layer **20** and does not extend over an edge of first print layer **20**. Second print layer **20** can be a next print layer **20** and, if additional print layers **20** are desired in stack **80** (step **130**), step **120** is repeated with a next print layer **20**, optionally with adhesive in step **115**, so that next print layer **20** is disposed on the immediately previous print layer **20** by transfer printing next print layer **20** from a print-layer source wafer **70** to the top of stack **80** of print layers **20**. Thus, methods of the present disclosure comprise successively transfer printing print layers **20** having successively smaller areas from a print-layer source wafer **70** onto stack **80** to increase the number of print layers **20** in stack **80**.

(87) Each functional layer **24** comprises an exposed conductive portion **25** that is not covered with a dielectric layer **22** of any of print layers **20**. In some embodiments, print layers **20** on a print-layer source wafer **70** are oriented in a common direction but are rotated during the transfer printing process, e.g., by a stamp on a motion-control platform for transfer printing, to orient print layers **20** in different directions, for example on different sides of stack **80**, to locate exposed conductive portions **25** on the different sides (edges) of stack **80**, simplifying the pattern of electrodes **32**.

(88) Each print layer **20** comprises a dielectric layer **22** and a functional layer **24** disposed on dielectric layer **22**. Each functional layer **24** comprises an exposed conductive portion **25** that is not covered with a dielectric layer **22** of any print layers **20** in stack **80**. Thus, second print layer **20** (e.g., next print layer **20**) has a smaller area than the first print layer (e.g., print layers **20** beneath the next print layer **20** in stack **80**). Each exposed conductive portion **25** is spatially separated from any other exposed conductive portion **25** in a direction orthogonal to stack **80**, e.g., each exposed conductive portion **25** is horizontally separated by a spatial separation **S** in horizontal direction **H** from every other exposed conductive portion **25**. Spatial separation **S** can be zero so that two exposed conductive portions **25** can abut or border on each other but cannot overlap, e.g., are nonoverlapping exposed conductive portions **25**.

(89) After stack **80** of print layers **20** is completed, in optional step **135**, as shown in FIGS. **1E** and **1F**, any adhesive **60**, if present, is removed, e.g., by exposure to an energetic plasma that can remove organic materials such as polymeric adhesives **60**, thus clearing exposed conductive portions **25**. Electrode layer **30** (an electrical conductor) is deposited in step **140** as shown in FIG. **1G**, for example by evaporation or sputtering a metal or other electrically conductive material as an unpatterned coating over stack **80**. The electrode layer **30** is patterned in step **150** to form electrodes **32**, for example by photolithographic methods and materials. Stacked electronic component **90** is then complete, for example as shown in FIGS. **1H** and **1I**.

(90) In some embodiments of the present disclosure, stack **80** of print layers **20** (e.g., stacked electronic component **90**) is disposed on a component substrate **10**, e.g., as shown in FIGS. **1E** and **1F**. In some embodiments, component substrate **10** is originally disposed on a component source wafer **76** (e.g., as shown in FIG. **20**) and the entire stacked electronic component **90** (e.g., comprising stack **80** of print layers **20** and component substrate **10**) is released from component source wafer **76** in step **160** and transfer printed (e.g., micro-transfer printed with a stamp in a

motion platform) from component source wafer **76** to a target substrate **50** provided in step **104** in step **170**, for example as shown in FIGS. **15** and **16B**.

(91) In some embodiments, component substrate **10** is a final substrate (e.g., is also a target substrate **50**) with other electrical components or circuits disposed thereon to make a final electronic system for an application. As shown in FIG. **18**, in some such embodiments, methods of making a stacked electronic component **90** comprise providing one or more print-layer source wafers **70** comprising print layers **20** in step **100** and as shown in FIGS. **1A-1D**, **19** and providing a target substrate **50** in step **104**. Optionally, a layer of adhesive **60** is disposed on target substrate **50** in step **106**. A first print layer **20** is transfer printed from one of print-layer source wafers **70** onto target substrate **50** in step **112**. Optionally, in step **115**, adhesive **60** is disposed on first print layer **20**. In step **120**, a second print layer **20** is transfer printed from a print-layer source wafer **70** (e.g., the same print-layer source wafer **70** that provided first print layer **20** or a different print-layer source wafer **70**) onto first print layer **20** to form a stack **80** of print layers **20**. Second print layer **20** has a smaller area over target substrate **50** than first print layer **20**. In embodiments, second print layer **20** is completely and entirely disposed directly over first print layer **20** and does not extend over an edge of first print layer **20**. Second print layer **20** can be a next print layer **20** and, if additional print layers **20** are desired in stack **80** (step **130**), step **120** is repeated with a next print layer **20** so that next print layer **20** is disposed on the immediately previous print layer **20** by transfer printing next print layer **20** from a print-layer source wafer **70** to the top of stack **80** of print layers **20**. Thus, methods of the present disclosure comprise successively transfer printing print layers **20** having successively smaller areas from a print-layer source wafer **70** onto stack **80** to increase the number of print layers **20** in stack **80**. Each functional layer **24** comprises an exposed conductive portion **25** that is not covered with a dielectric layer **22** of any of print layers **20**. In some embodiments, print layers **20** on a print-layer source wafer **70** are oriented in a common direction but are rotated during the transfer printing process, e.g., by a stamp on a motion-control platform for transfer printing, to orient print layers **20** in different directions, for example on different sides of stack **80**, to locate exposed conductive portions **25** on the different sides of stack **80**, simplifying the pattern of electrodes **32**. Once all of print layers **20** are disposed in stack **80** (step **130**), in step **135** any adhesive **60**, if present, is removed, e.g., by exposure to an energetic plasma that can remove organic materials such as polymeric adhesives **60**, thus clearing exposed conductive portions **25**, for example as shown in FIG. **13**. Electrode layer **30** (an electrical conductor) is deposited in step **140**, for example by evaporation or sputtering a metal or other electrically conductive material as an unpatterned coating over stack **80**, for example as shown in FIG. **1G**. Electrode layer **30** is patterned in step **150** to form electrodes **32**, for example by photolithographic methods and materials. Stacked electronic component **90** is then complete and disposed on target substrate **50** and can be electrically interconnected with other electronic or optoelectronic components in a system.

(92) In some methods of the present disclosure, stack **80** is coated with unpatterned insulating layer **40** and patterned to form vias **44** in patterned insulating layer **42** that expose exposed conductive portions **25**. Some methods of the present disclosure comprise coating stack **80** with an electrode layer **30** and patterning electrode layer **30** to define one or more electrodes **32**, each electrode **32** in electrical contact with an exclusive subset of exposed conductive portions **25**. According to some methods, patterning electrode layer **30** comprises forming functional-layer connectors **34** that each electrically connect exposed conductive portions **25** of at least two print layers **20** and are electrically separate from electrodes **32** so that electrodes **32** electrically connect fewer than all of exposed conductive portions **25**.

(93) In embodiments according to FIG. **21**, dielectric layer **22** of print layers **20** can have a rectangular shape with a tab extending from the rectangular area and exposed conductive portion **25** (e.g., exposed conductive portions **25A**, **25B**) can be disposed on the tabs. Such an arrangement can simplify the structure of top dielectric layer **26** or electrodes **32**, particularly if unpatterned

layers of adhesive **60** are disposed, for example by spray coating. The tabs can provide a useful masking effect for material deposition.

(94) Embodiments of the present disclosure describe passive stacked electronic components **90** (e.g., micro-assembled capacitor **97**, micro-assembled resistor **98**, micro-assembled inductor **99**, or micro-assembled antenna **96**) comprising stacks **80** of passive print layers **20** for which the term electronic includes an electrical conductor used in an electronic circuit or system and does not imply that stacked electronic components **90** are active components, for example comprising a transistor.

(95) According to some embodiments of the present disclosure and as shown in active print layer **20** of FIG. **22**, an active stacked electronic component **95** comprises a stack **80** of active print layers **20**. Each print layer **20** in stack **80** has an area less than any print layers **20** beneath print layer **20** in stack **80**. Each print layer **20** comprises a dielectric layer **22** and an epitaxial layer **23** (that is a functional layer **24**) disposed on or in dielectric layer **22**. Epitaxial layer **23** comprises an exposed conductive portion **25** that is not covered with a dielectric layer **22** of any of active print layers **20**, and each exposed conductive portion **25** is nonoverlapping with any other exposed conductive portion **25**. Epitaxial layer **23** can be a semiconductor substrate with a process surface and can comprise an active circuit **29** formed in or on the process surface, for example comprising transistors or diodes. A top dielectric layer **26** can be disposed over epitaxial layer **23**. Dielectric layer **22** can be a layer of adhesive **60** on which epitaxial layer **23** (e.g., a semiconductor substrate) is disposed (e.g., by micro-transfer printing a semiconductor substrate onto layer of adhesive **60**) when functional layer **24** is epitaxial layer **23**. A patterned electrode layer **30** is coated on at least a portion of stack **80**, patterned electrode layer **30** defining one or more electrodes **32**. Each electrode **32** is in electrical contact with an exclusive subset of exposed conductive portions **25**. In some embodiments, passive print layers **20** are interspersed between active print layers **20** in stack **80**, for example providing both active stacked electronic components **90** and passive stacked electronic components **90** in a common stack **80**.

(96) Active circuit **29** can comprise electronic circuitry, structures, and materials, for example electrical conductors, vias, doped semiconductors formed using lithographic processes. The circuits can include insulating layers and structures such as silicon dioxide, nitride, and passivation layers and functional layers or structures including wires or circuit electrodes made of aluminum, titanium, silver, or gold that form an electronic circuit. Useable methods and materials for making electronic circuits are known in the integrated circuit arts.

(97) In some embodiments of the present disclosure, active stacked electronic components **95** comprise active print layers **20** (e.g., comprising an epitaxial layer **23** with active elements such as transistors) and passive print layers **20** (e.g., forming capacitor plates, resistor wires, or inductor wires). Moreover, a single stack **80** can comprise different types of active or passive functional layers **24** and can make multiple different electronic components in the common stack **80**. For example, in a single stack **80** some print layers **20** can provide resistors, some print layers **20** can provide capacitors, and some print layers **20** can provide active elements to form a functionally heterogeneous stacked electronic component **90** electrically connected with functional-layer connectors **34**. In general, functional-layer connectors **34** can connect any print layers **20** in stack **80** in any order in series or in parallel to form the desired stacked electrical component **90**.

(98) According to some embodiments of the present disclosure, and as shown in FIG. **23**, target substrate **50** (or component substrate **10**) is a semiconductor substrate with a target substrate active circuit **52** (or component active circuit **52**) and target substrate contact pads **56** formed in or disposed thereon and stacked electronic component **90** is disposed over target substrate active circuit **52** and electrically connected to target substrate active circuit **52** with target substrate electrodes **54** (e.g., corresponding to component electrodes **36**) and target substrate contact pads **56**, to form a three-dimensional integrated circuit. This structure enables an efficient and highly integrated electronic system incorporating active and passive electronic components.



(99) The use of transfer printing (e.g., micro-transfer printing) reduces construction costs. If print layers **20** are serially constructed using photolithography to form stack **80**, the materials in each print layer **20** must be successively deposited and patterned, relatively slowly and at significant expense. In contrast, forming all of print layers **20** on print-layer source wafer **70** in a common step greatly reduces the amount of photolithographic processing necessary. Since the transfer printing steps can be done massively in parallel (e.g., **10,000** to **100,000** print layers **20** per transfer, where each transfer takes only a fraction of a minute), costs and processing time are much reduced.

(100) Stacked electronic components **90** can be very small (e.g., having a length and a width no greater than 200 microns (for example, no greater than 100 microns, no greater than 50 microns, no greater than 20 microns, or no greater than 10 microns) and a thickness no greater than 50 microns (for example no greater than 30 microns, no greater than 20 microns, no greater than 10 microns, or no greater than five microns). Each print layer **20** can be very thin, for example one micron, two microns, or five microns thick. Despite the small size, the functional performance of stack **80** of stacked electronic components **90** is increased by an approximate multiple of the number of print layers **20** in stack **80**. In certain embodiments, stacked electronic components **90** of the present disclosure can be micro-transfer printed. Micro-transfer printing can transfer very small components (e.g., integrated circuit chiplets) from a source wafer to a target substrate **50**. Other methods, such as pick-and-place or surface-mount techniques cannot transfer such small stacked electronic components **90**. Because the stacked electronic components **90** are relatively small, they can be disposed directly on a semiconductor die or adjacent to an unpackaged semiconductor die. The semiconductor die can be, for example, an integrated circuit such as a CMOS circuit.

(101) Target substrates **50** can be any suitable substrate on which an electronic or opto-electronic circuit can be constructed or disposed and can be a glass, polymer, ceramic, quartz, or semiconductor substrate. Target substrate **50** can be a semiconductor wafer and circuits can be disposed on target substrate **50** by transfer printing or can be natively formed in situ, for example as CMOS, digital, mixed signal, or analog circuits in an epitaxial layer of a semiconductor wafer. Electrical target substrate contact pads **56** and target substrate electrodes **54** can be provided on target substrate **50** that are electrically connected to electrodes **32** or component electrodes **36**, for example by photolithographic or printed circuit board methods and materials.

(102) In some embodiments of the present disclosure, a component source wafer **76** or print-layer source wafer **70** and sacrificial portion **72** include various materials. In some embodiments, component source wafer **76** or print-layer source wafer **70** is anisotropically etchable (for example silicon {1 1 1}) and each sacrificial portion **72** is a designated portion of component source wafer **76** or print-layer source wafer **70**. In some embodiments, each sacrificial portion **72** comprises sacrificial material (e.g., silicon dioxide) that is differentially etchable from component source wafer **76** or print-layer source wafer **70**. In some embodiments, sacrificial portion **72** is etched forming a gap between print layer **20** and print-layer source wafer **70** or between stack **80** and component source wafer **76** made by etching sacrificial portion **72**, for example with a wet etchant such as TMAH or KOH.

(103) According to various embodiments of the present invention, native source wafers can be provided with print layers **20**, sacrificial portion **72**, component substrates **10**, and component tethers **18** or print-layer tethers **28** already formed, or they can be constructed as part of a process in accordance with some embodiments of the present invention.

(104) Target substrates **50**, print-layer source wafers **70** (print-layer source substrates), component source wafers **76**, transfer-print stamps, and motion control platforms for micro-transfer printing can be made separately and at different times or in different temporal orders or locations and provided in various process states.

(105) For a discussion of micro-transfer printing techniques applicable to (e.g., adaptable to or combinable with) methods disclosed herein see U.S. Pat. Nos. 8,722,458, 7,622,367 and 8,506,867. Additional details useful in understanding and performing aspects of the present disclosure are

described in U.S. Patent Application Ser. No. 62/148,603 filed Apr. 16, 2015, entitled Micro Assembled Micro LED Displays and Lighting Elements and in U.S. Patent Application Ser. No. 62/055,472 filed Sep. 25, 2014, entitled Compound Micro-Assembly Strategies and Devices, the disclosure of each of which is hereby incorporated herein in its entirety by reference.

(106) Tethers that are usable with, adaptable for use in, or combinable with tethers disclosed herein are discussed in U.S. Patent Publication No. 2019/0385885 filed Jun. 14, 2018, entitled Multi-Layer Tethers for Micro-Transfer-Printing, and U.S. Pat. No. 10,714,374 filed May 9, 2019, entitled High-Precision Printed Structures, whose contents are incorporated herein by reference.

(107) As is understood by those skilled in the art, the terms “over” and “under”, “above” and “below”, “top” and “bottom” are relative terms and can be interchanged in reference to different orientations of the layers, elements, and substrates included in the present invention. For example, a first layer on a second layer, in some implementations means a first layer directly on and in contact with a second layer. In other implementations a first layer on a second layer includes a first layer and a second layer with another layer therebetween.

(108) Throughout the description, where apparatus and systems are described as having, including, or comprising specific components, or where processes and methods are described as having, including, or comprising specific steps, it is contemplated that, additionally, there are apparatus, and systems of the disclosed technology that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the disclosed technology that consist essentially of, or consist of, the recited processing steps.

(109) It should be understood that the order of steps or order for performing certain action is immaterial so long as operability is maintained. Moreover, two or more steps or actions in some circumstances can be conducted simultaneously.

(110) Having expressly described certain embodiments, it will now become apparent to one of skill in the art that other embodiments incorporating the concepts of the disclosure may be used.

Therefore, the claimed invention should not be limited to the described embodiments, but rather should be limited only by the spirit and scope of the following claims.

## PARTS LIST

(111) D stack direction/vertical direction H horizontal direction S spatial separation X x direction/x dimension Y y direction/Y dimension **10** component substrate **12** alignment structure **14** component functional layer **16** component dielectric layer **18** component tether **20** print layer **22** dielectric layer/bottom dielectric layer/dielectric substrate **23** epitaxial layer **24**, **24A**, **24B** functional layer **25**, **25A**, **25B** exposed conductive portion **26** dielectric layer/top dielectric layer **28** print-layer tether **29** active circuit **30** electrode layer **32**, **32A**, **32B** electrode **34** functional-layer connector **36** component electrode **40** unpatterned insulating layer **42** patterned insulating layer **44** via **50** target substrate **52** target substrate active circuit/component active circuit **54** target substrate electrode **56** target substrate contact pad **60** adhesive **70** print-layer source wafer **72** sacrificial portion **74** anchor **76** component source wafer/component source substrate **80** stack **90** stacked electronic component **95** stacked active electronic component/micro-assembled integrated circuit **96** stacked passive electronic component/micro-assembled antenna **97** stacked passive electronic component/micro-assembled capacitor **98** stacked passive electronic component/micro-assembled resistor **99** stacked passive electronic component/micro-assembled inductor **100** provide print-layer source wafer step **102** provide component source wafer step **104** provide target substrate step **106** optional pattern adhesive step **110** print first layer onto component wafer step **112** print first layer onto target substrate step **115** optional apply adhesive step **120** print next layer onto previous layer step **130** all layers printed step **135** optional pattern adhesive step **140** deposit conductor step **150** pattern electrodes step **160** release component step **170** print component step

## Claims

1. A method of making a stacked electronic component, comprising: providing one or more print-layer source wafers each comprising multiple print layers, each of the print layers comprising a dielectric layer and a functional layer disposed on the dielectric layer, each functional layer comprising an exposed conductive portion that is not covered with a dielectric layer of any of the print layers; providing a component source substrate; transfer printing a first print layer from one of the one or more print-layer source wafers onto the component source substrate; transfer printing a second print layer from one of the one or more print-layer source wafers onto the first print layer to form a stack of print layers; transfer printing a third print layer from one of the one or more print-layer source wafers onto the second print layer thereby increasing the number of print layers in the stack of print layers; after printing the third print layer onto the stack of print layers, coating the stack of print layers with an electrode layer; and patterning the electrode layer to define one or more electrodes, the one or more electrodes in electrical contact with each exposed conductive portion of an exclusive subset of the print layers in the stack to make the stacked electronic component, wherein the second print layer has a smaller area than the first print layer and the third print layer has a smaller area than the second print layer and each exposed conductive portion of the print layers in the stack is nonoverlapping with any other exposed conductive portion.
  2. The method of claim 1, further comprising successively transfer printing print layers having successively smaller areas from a print-layer source wafer of the one or more print-layer source wafers onto the stack to increase the number of print layers in the stack, wherein each functional layer comprises the exposed conductive portion that is not covered with the dielectric layer of any of the print layers.
  3. The method of claim 1, wherein the one or more print-layer source wafers are a single, common print-layer source wafer.
  4. The method of claim 1, further comprising rotating the second print layer with respect to the first print layer while transfer printing the second print layer onto or over the first print layer.
  5. The method of claim 1, wherein the method further comprises providing a target substrate and transfer printing the stacked electronic component to the target substrate.
  6. The method of claim 1, further comprising coating the electrode layer with an insulating layer and forming vias in the insulating layer to expose the exposed conductive portions.
  7. The method of claim 1, wherein patterning the electrode layer comprises forming a functional-layer connector that electrically connects the exposed conductive portions of at least two print layers in the stack and is physically and spatially separate from the one or more electrodes so that the one or more electrodes electrically connect fewer than all of the exposed conductive portions in the stack.
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